

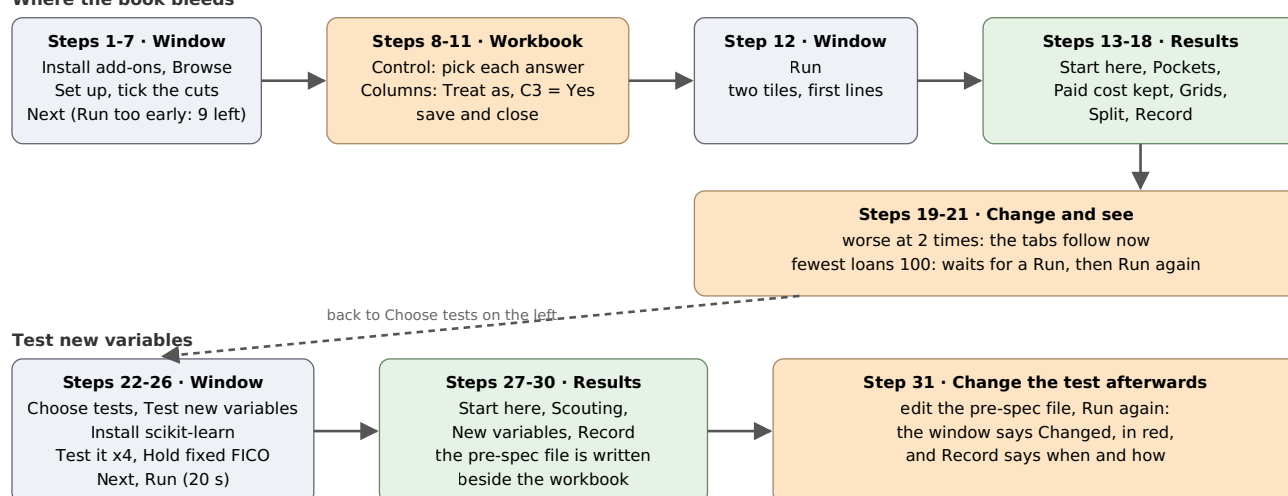
PocketBook, step by step: finding where a loan book bleeds, and testing a new variable

Who this is for: an analyst using PocketBook for the first time. You need the loan file from the bank and Excel. You don't type any commands.

What you'll have at the end: a workbook beside the loan file that shows which groups of loans lose more than their share, how much, and whether it could be chance. Then a second kind of run: whether a new column (here, revolving debt) really tells good loans from bad, checked on loans it was never found on.

Walked on: 27 Sep 2026, on a made-up book of 8,000 loans with known answers planted in it. Every picture below is the real screen from that walk.

Where the book bleeds



Grey: the PocketBook window. Peach: where you answer (Excel, or the pre-spec file). Green: where you read the results.
Nothing is typed at a command line. Step 32 is what a Run that stops looks like.

The route: the window, the workbook, the window again, then the results, for each kind of run.

Words you'll meet

Each is shown the first time on screen. This is what they mean.

On screen	What it means
Extract	The loan file from the bank: one row per loan, as a .csv or .xlsx. PocketBook never changes it.
Workbook	The Excel file PocketBook writes beside the extract. You answer questions in it, and the results land in it.
Band	A range of a number column, like a FICO score from 496 to 653. Cut into bands means split the column into ranges.
Segment	A group from a category column, like the Broker channel.
Pocket	One band crossed with one segment: FICO 496 to 653 and Broker. The thing PocketBook judges.
Grid	Every band of one column against every segment of another: one table of pockets.

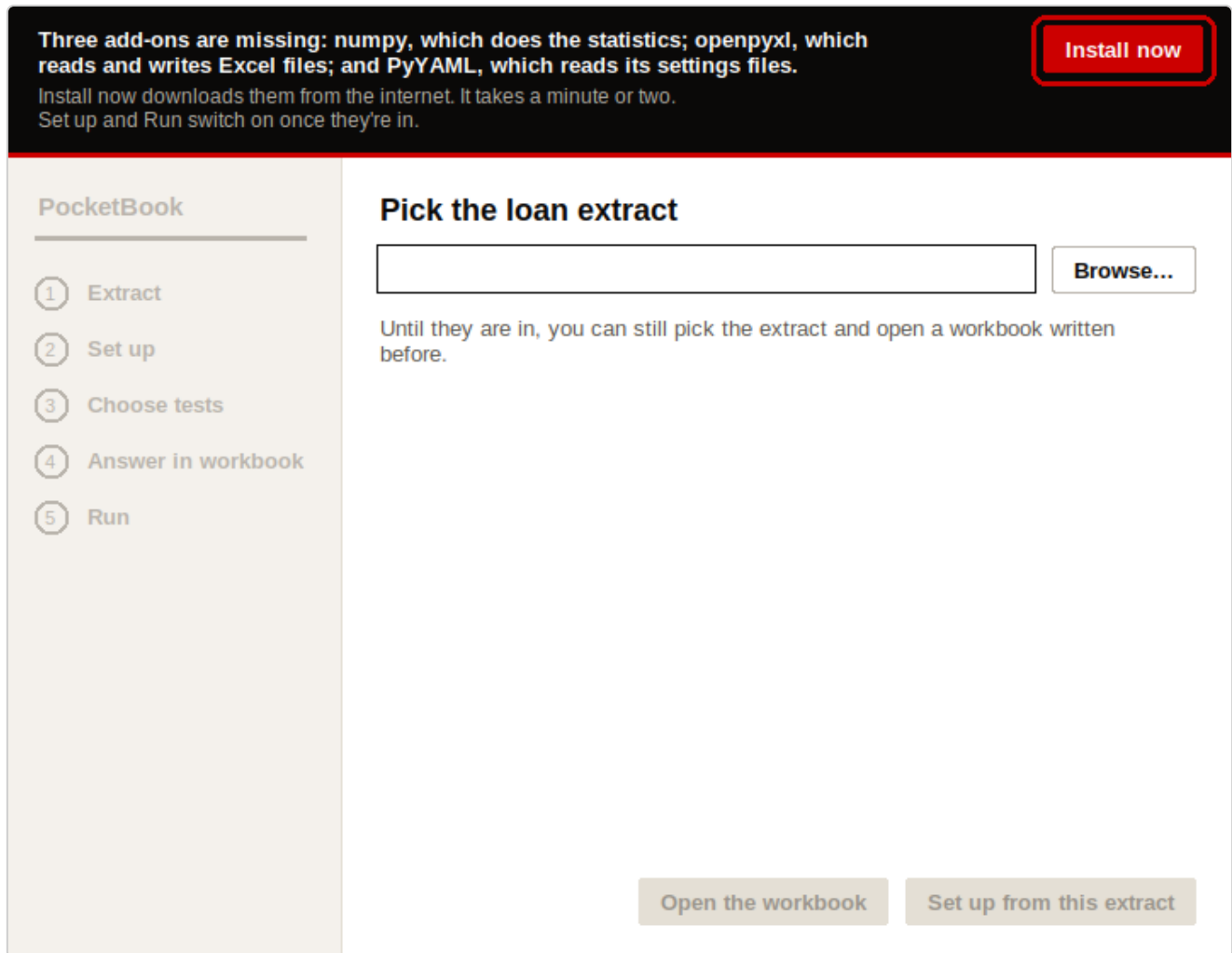
On screen	What it means
GCO	Gross charge-offs: the dollars written off on loans that went bad. On screen as Charge-offs .
RANR	What the bank kept after losses: interest and fees, less the cost of funds, less charge-offs. On screen as Kept after losses .
Worse at 1.34x	A pocket counts as worse when it goes bad at least 1.34 times as often as the loans it is compared with.
p-value	How likely a gap this big is by chance alone. Under 5% means it is unlikely to be chance.
Material	Big enough in dollars to matter: at or above the line you pick on Control.
Pre-spec	A short file that fixes, before the test, exactly what will be tested. It stops anyone moving the goalposts after seeing the answer.
Development loans / held back	The loans are split by date. The first 70% are where an idea is found (development). The last 30% are held back and only used to check it.
Scouting	The step that looks through the columns you ticked, on the development loans, and proposes which are worth testing.

Part A · Open PocketBook

Step 1 · Double-click PocketBook.pyw

The window opens with five steps down the left. The first time, a black bar may say some add-ons are missing. **Add-ons** are free pieces of software PocketBook needs to do its sums and read Excel.

Press **Install now**.



Correct screen: the bar names what is missing and what each one does. **Set up** is grey until they're in.

Step 2 · Pick the extract

Press **Browse...**, find the loan file, and press **Open**.

Three add-ons are missing: numpy, which does the statistics; openpyxl, which reads and writes Excel files; and PyYAML, which reads its settings files.

Install now

Install now downloads them from the internet. It takes a minute or two.
Set up and Run switch on once they're in.

PocketBook

Pick the loan extract

1 Directory: /home/user/SATC/.claude/worktrees/agent-ae4911b224660d84f

2

3

4

5

.claude	cashew-institute	credit-suite
.claude-plugin	cfpb-mortgage-monitor	desk
.github	client-documents	docs
bls-laus-county-monitor	consumer-credit-research	edgar-crit-cl
bureau-credit-risk-dashboard	cowork-plugin	fdic-peer-mo
canon	credit-review-os	fred-credit-ri

File name: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-

Files of type: Extracts (*.csv,*.xlsx)

Open Cancel

Open the workbook

Set up from this extract

Step 3 · Wait for the install

It takes a minute or two (a few seconds if the add-ons were fetched before).

PocketBook

1 Extract

Pick the loan file

2 Set up

3 Choose tests

4 Answer in workbook

5 Run

Pick the loan extract

One row per loan, as a .csv or .xlsx. Set up reads it and guesses what each column means. The workbook is written beside it once you've chosen the tests. The extract itself is never changed.

-4db589c3cc31/scratchpad/walk27/book2/Consumer book Q3.csv

Browse...

▼ How Set up recognises columns

A number column with this many values or fewer is a category

12 values ▼

A text column with more values than this is too fine to cut by

50 values ▼

Installed numpy, openpyxl and PyYAML. Everything PocketBook needs is here.

Set up from this extract

Correct screen: the black bar has gone. A line says what was installed. **Set up from this extract** is red, so you can press it. The two settings under How Set up recognises columns can be left as they are.

Part B · Where the book bleeds

Step 4 · Press Set up from this extract

PocketBook reads the file and guesses what each column is. Nothing is written yet.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**
10 columns read
- 3 Choose tests**
Where the book bleeds
- 4 Answer in workbook
- 5 Run

What are you running?

Where the book bleeds

Test new variables

Column	What it is	Cut into bands	Segment by	Split by
LOAN_NBR	Loan number (key)			
FICO	FICO score	✓		
CHANNEL	Category - 3 values		✓	
ORIG_BAL	Booked amount	✓		
BAD_FLAG	Outcome (yes/no)			· every measure
GCO_AMT	GCO dollars			· every measure
RANR_AMT	RANR dollars			· every measure
ASSET_CLASS	Category - 4 values		✓	
REV_DEBT	Amount or number	✓		
ORIG_DATE	Origination date			

This will run: 3 band columns × 2 segment columns = 6 grids, five measures each.

GCO dollars: what a loan charged off, in dollars.
RANR dollars: what we kept from a loan after its losses, in dollars.
A grid: every band of one column against every segment of another.
Five measures: bad loans, bad dollars, charge-offs, earned before and kept after losses.

Saved to Control, read-only.

Next: answer in the workbook →

Correct screen: the left side shows the file name and "8,000 loans", then "10 columns read". The table lists every column with what PocketBook thinks it is. **Where the book bleeds** is picked at the top right.

The boxes decide what is cut:

- **Cut into bands:** number columns to split into ranges (FICO, ORIG_BAL, REV_DEBT are ticked).
- **Segment by:** category columns (CHANNEL, ASSET_CLASS).
- **Split by:** at most one number column, used to cut every pocket in half at its own middle value.
- The outcome and dollar columns say **every measure**: they are what is measured, not what is cut.

Under the shaded box, a line for each word you meet here first: GCO dollars, RANR dollars, a grid, and the five measures.

The made-up book has three number columns you can cut. Your bank's file may have more.

Step 5 · Tick Split by on REV_DEBT

Press the round button in the **Split by** column on the REV_DEBT row.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**
10 columns read
- 3 Choose tests**
Where the book bleeds
- ④ **Answer in workbook**
- ⑤ **Run**

What are you running?

Where the book bleeds | Test new variables

Column	What it is	Cut into bands	Segment by	Split by
LOAN_NBR	Loan number (key)			
FICO	FICO score	✓		
CHANNEL	Category · 3 values		✓	
ORIG_BAL	Booked amount	✓		
BAD_FLAG	Outcome (yes/no)	· every measure		
GCO_AMT	GCO dollars	· every measure		
RANR_AMT	RANR dollars	· every measure		
ASSET_CLASS	Category · 4 values		✓	
REV_DEBT	Amount or number	☐		☒
ORIG_DATE	Origination date			

This will run: 2 band columns × 2 segment columns = 4 grids, five measures each; split by REV_DEBT adds 4 more.

GCO dollars: what a loan charged off, in dollars.
RANR dollars: what we kept from a loan after its losses, in dollars.
A grid: every band of one column against every segment of another.
Five measures: bad loans, bad dollars, charge-offs, earned before and kept after losses.

Saved to Control, read-only.

Next: answer in the workbook →

Correct screen: REV_DEBT's **Cut into bands** box clears by itself (a column can't be both). The shaded box under the table says what will run: 2 band columns × 2 segment columns = 4 grids ... split by REV_DEBT adds 4 more.

Step 6 · Press Next: answer in the workbook

The workbook is written beside the extract: Consumer book Q3 - PocketBook.xlsx.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**
10 columns read
- ✓ **Choose tests**
Where the book bleeds
- 4 Answer in workbook**
Fill in the shaded cells
- 5 **Run**

Answer in the workbook

Consumer book Q3 - PocketBook.xlsx is written beside the extract. Fill in the shaded cells, save and close it, then press Run.

Worked out from this extract: fewest loans 65; worse at 1.34x; better at 0.75x. Each sits beside its setting on Control, for you to take or not.

worse at 1.34x: losing 1.34 times as much as the rest counts as worse.
better at 0.75x: losing 0.75 times as much as the rest counts as better.

Next: 9 answers needed before Run, on Control and Columns. Fill in the shaded cells, save, close, and press Run.

Open the workbook

Run

Correct screen: step 4 on the left is red. The shaded box gives three values PocketBook worked out from this book (fewest loans in a pocket, how much worse, how much better), with a line under it on what worse at and better at mean. They are only suggestions, and each sits beside its question on the Control tab. Under the box: **9 answers needed before Run, on Control and Columns**. Start here and the Run itself (Step 7) give the same count.

Step 7 · (What happens if you press Run too early)

If you press **Run** now, nothing runs. The window lists every answer still needed, with the tab and cell.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**
10 columns read
- ✓ **Choose tests**
Where the book bleeds
- ! Answer in workbook**
9 left
- 5 **Run**

9 answers needed before Run

Each one opens the workbook at its cell. Nothing ran; your answers so far are kept.

CONTROL C15	How much worse than its comparison a pocket must be	Open at C15
CONTROL C16	How much better than its comparison a pocket must be	Open at C16
CONTROL C17	How far profit must move before it counts as more or less	Open at C17
CONTROL C18	How sure a difference must be before it counts	Open at C18
CONTROL C19	Smallest excess loss worth reporting	Open at C19
CONTROL C20	What a pocket is judged against	Open at C20
CONTROL C24	Fewest loans in a pocket for the usual test (smaller ones get the exact test)	Open at C24

Open the workbook
Run

Correct screen: "9 answers needed before Run", top to bottom as the workbook shows them. Each has an **Open at** button that opens the workbook on that cell. Step 4 on the left says "9 left". Your answers so far are kept.

Step 8 · Open the workbook and go to Control

Press **Open the workbook**. The **Control** tab holds the calls only you can make. Every pink cell needs an answer.

Control

The professional calls. The top block changes results now; the second waits for a Run.

How this tab works
Changes now
Needs a Run
Your answer
Worked out
Chosen in the launcher
Materiality levels

The result tabs read these answers with formulas. Change one and every reading, dollar figure, colour and verdict follows at once. The order of rows stays as the last Run left it. These decide which pockets exist and which test each one gets, so they take effect when you press Run again. Status says when an answer differs from what the last Run used. Pick from the list, or type a number under Or your own: a number there wins. A shaded answer still needs one. Comes to shows what the answer amounts to now. Fewest loans, worse at and better at each have a value worked out from this extract, shown beside the setting. It is never picked for you. What you're running and how the pockets are cut. Change them in the launcher's Choose tests, then press Next. They are shown here so a reviewer sees them. For each level: its dollar line, how many pockets have charge-offs above their share that reach it, and their part of all such dollars. It follows your answer live.

Changes now	Needs a Run	Still needs an answer
Changes now - the result tabs follow as you change these		
Setting	Your answer to	Or your own
How much worse than its comparison a pocket must be		
How much better than its comparison a pocket must be		
How far profit must move before it counts as more or less		
How sure a difference must be before it counts		
Smallest excess loss worth reporting		
What a pocket is judged against		
To Needs a Run - takes effect when you press Run again		
Setting	Your answer to	Or your own
Fewest loans in a pocket for the usual test (smaller ones get the exact test)		
Fewest loans with a loss before a loss rate is tested		
How many bands a number column is cut into	5 bands (recommended)	
Where the band edges fall	Each band holds about the same number of loans (recommended)	
How often a real gap should be caught	80% of the time (recommended)	
Allowing for testing many pockets at once	Hold down the share of false finds (Benjamini-Hochberg) (recommended)	
Chosen in the launcher - read-only here		
What are you running?	Where the book bleeds	
Cut into bands	FICO, ORIG_BAL	
Segment by	CHANNEL, ASSET_CLASS	
Split every pocket by	REV_DEBT	
A number column with this many values or fewer is a category	12 values	
A text column with more values than this is too fine to cut by	50 values	

To change these, go back to Choose tests in the launcher and press Next.

At the right, under **Worked out from the loans**, are the values PocketBook suggests for this book:

[illegible]

The top block (**Changes now**) takes effect in the result tabs as soon as you change it. The second block (**Needs a Run**) takes effect the next time you press Run.

Step 9 · Answer Control

Pick from each cell's dropdown. On the walk, reading the suggestions:

Question	Answer picked	Why
How much worse than its comparison a pocket must be	The smallest significant gap in a typical pocket (suggested)	comes to 1.34 times, worked out from this book
How much better ...	The smallest significant gap in a typical pocket (suggested)	0.75 times
How far profit must move before it counts	Each pocket's own test (suggested)	
How sure a difference must be before it counts	95% sure	
Smallest excess loss worth reporting	1% of the book's total losses	comes to \$107,354 once the Run has added up the book's losses
What a pocket is judged against	The rest of its band	
Fewest loans in a pocket for the usual test	Enough for 5 expected losses (suggested)	65 loans
Fewest loans with a loss before a loss rate is tested	10 losses	

The rest already hold a recommended answer.

Control

The professional code. The top block changes results now. The second waits for a Run.

How this tab works

Changes now

Needs a Run

Your answer

Worked out

Chosen in the launcher

Materiality levels

The result tabs read these answers with formulas. Change one and every reading, dollar figure, colour and verdict follows at once. The order of rows stays as the last Run left it.

These decide which pockets exist and which test each one gets, so they take effect when you press Run again. Status says when an answer differs from what the last Run used.

Pick from the list, or type a number under Or your own: a number there wins. A shaded answer still needs one. Comes to shows what the answer amounts to now.

Freest loans, worse at and better at each have a value worked out from this extract, shown beside the setting. It is never picked for you.

What you're running and how the pockets are cut. Change them in the launcher's Choose tests, then press Next. They are shown here so a reviewer sees them.

For each level: its dollar line, how many pockets have charge-offs above their share that reach it, and their part of all such dollars. It follows your answer live.

Changes now	Needs a Run	Still needs an answer
Changes now: the result tabs follow as you change these Setting How much worse than its comparison a pocket must be How much better than its comparison a pocket must be How far profit must move before it counts as more or less How far profit must move before it counts as more or less How sure a difference must be before it counts Smallest excess loss worth reporting What a pocket is judged against	Your answer The smallest significant gap in a typical pocket (suggested) The smallest significant gap in a typical pocket (suggested) Each pocket's own test (suggested) 95% sure 1% of the book's total losses The rest of its band	Or your own 1.34x 0.75x Last Run used suggested: 1.34x, from this extract suggested: 0.75x, from this extract
Needs a Run: takes effect when you press Run again Setting Fewest loans in a pocket for the usual test (smaller ones get the exact test) Fewest loans with a loss before a loss rate is tested How many bands a number column is cut into Where the band edges fall How often a real gap should be caught Allowing for testing many pockets at once	Your answer to Enough for 5 expected losses (suggested) 10 losses 5 bands (recommended) Each band holds about the same number of loans (recommended) 80% of the time (recommended) Hold down the share of false finds (Benjamin-Hochberg) (recommended)	Or your own Last Run used Status Worked out from the loans suggested: 65, from this extract
Chosen in the launcher: read-only here What are you running? Cut into bands Segment by Split every pocket by A number column with this many values or fewer is a category A text column with more values than this is too fine to cut by	Where the book bleeds FICO, ORIG_BAL CHANNEL, ASSET_CLASS REV, DEBT 12 values 50 values	To change these, go back to Choose tests in the launcher and press Next.

What

Line

0.5%

1%

2%

5%

10%

Cut

A pr

Correct screen: no pink cell left on Control. **Comes to** already shows **1.34x** and **0.75x** beside the two suggestions, before any Run.

Step 10 · Answer Columns

On the **Columns** tab:

1. Read each row's **What it is** and **Why we think so**. Fix any that's wrong.
2. Answer each **Odd value** under **Treat as**. On the walk: **FICO -9999 on 160 loans** is a code for "no score", so **Missing**. **RANR negative on 595 loans** is real (a loan that charged off loses money), so **Real**.
3. Set **Checked every column?** (C3) to **Yes**.

Columns What each column means, what looks odd in it, and what PocketBook remembers about it.

Checked every column: Run won't start until this is Yes.

How this tab works

What it is Set up guesses each column's meaning from its name and values; the reason is beside it. Fix any that's wrong. Confirmed meanings are remembered for the next extract.

Odd values Values that may be codes rather than real numbers: one value far more often than any other, or negatives in a column that's mostly positive. Missing is treated as blank and counted.

Treat as Answer Real or Missing. One left blank is used as recorded.

Band edges Blank means Control's setting. Type edges as 620; 680; 740, or every 20. Look shows what each would cut, live.

Remembered Set Forget? to Yes to drop what was learned about a column at the next Run. It then waits here for you to confirm it again. Answers marked to take effect at the next Run.

Column	Samples	What it is to	Why we think so	Blank	Odd values	Treat as to	Band edges to	Remembered
LOAN_NBR	L0000000, L0000001, L0000002	Loan number (key)	name contains 'loan'; a different value on every row	0%				N
FICO	-9999, 713, 716	FICO score	name contains 'fico'; 98% of values between 300 and 850	0%	-9999 on 160 loans			N
CHANNEL	Online, Broker, Branch	Category	3 different values	0%				N
ORIG_BAL	7655.75, 27999.47, 39684.43	Booked amount	name contains 'bal'; positive on every loan	0%				N
BAD_FLAG	0, 2, 1	Outcome (yes/no)	name contains 'bad'; 0 and 1 (7.7% are 1)	0%				N
GCO_AMT	0.0, #N/A, 27620.62	GCO dollars	name contains 'gco'; zero on 92% of loans, never negative	0%				N
RANR_AMT	3017.35, 3928.24, 5816.98	RANR dollars	name contains 'ranr'; numbers	0%	Negative on 595 loans			N
ASSET_CLASS	1, 2, 4	Category	4 different values	0%				N
REV_DEBT	15868.86, 12884.7, 8920.97	Amount or number	numbers with 7,889 values	0%				N
ORIG_DATE	2021-11-18, 2024-06-24, 2023-07-1	Origination date	name contains 'orig'; dates	0%				N

Add a column: one divided by another

Shows here and on Look, and in Choose tests, once you press Set up again. Blank where the bottom is zero or blank.

New column name to	Top (divided)	Bottom (divided by)	What it makes

Step 11 · Save and close the workbook

While it is open in Excel, the window shows a pink bar and **Run** stays grey, because Run can't read answers Excel is still holding.

PocketBook

✓ **Extract**
Consumer book Q3.csv - 8,000 loans

✓ **Set up**
10 columns read

✓ **Choose tests**
Where the book bleeds

4 **Answer in workbook**
Fill in the shaded cells

5 **Run**

Answer in the workbook

Consumer book Q3 - PocketBook.xlsx is written beside the extract. Fill in the shaded cells, save and close it, then press Run.

The workbook is open in Excel.
Save and close it first, so Run reads your latest answers.

Worked out from this extract: fewest loans 65; worse at 1.34x; better at 0.75x. Each sits beside its setting on Control, for you to take or not.
worse at 1.34x: losing 1.34 times as much as the rest counts as worse.
better at 0.75x: losing 0.75 times as much as the rest counts as better.

Next: 9 answers needed before Run, on Control and Columns. Fill in the shaded cells, save, close, and press Run.

Open the workbook

Run

Correct screen after closing: the pink bar goes by itself within a couple of seconds, and **Run** turns red.

Step 12 · Press Run

On 8,000 loans it took 9 seconds.

PocketBook

✓ Extract

Consumer book Q3.csv · 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Where the book bleeds

✓ Answer in workbook

All answered

✓ Run

12:10 · 8 seconds

Run finished. Results are in the workbook.

Pockets worse and material

4

charge-offs, of 81

Charge-offs above their share

\$3.09M

in those 4 pockets

Open Consumer book Q3 - PocketBook.xlsx: start with Pockets. Split by REV_DEBT: each pocket halved at its own median. See the Split tab, and Pockets split by REV_DEBT. REV_DEBT isn't cut on its own while it splits.

Run again

Open at Start here

Correct screen: "Run finished." Two tiles: how many pockets are worse on charge-offs and material, and the charge-off dollars above their share in them. Under them, the Run's first two lines: where to start reading, and what splits the pockets. PocketBook also checks that every grid adds back up to the book. If one didn't, the Run would stop there and say so (Step 32); Record keeps how many checks there were.

Step 13 · Read Start here

Press **Open at Start here**.

PocketBook

Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 12:10

How this tab works

Where things stand

What is left before Run, counted live from Control and Columns as you fill them in.

What the last Run found

Pockets worse and material at Control's lines, and the five largest.

The tabs

Red tabs you fill in; black tabs hold results; grey tabs are the record.

WHERE THINGS STAND

Answers needed before Run

Odd values to answer

Changes waiting for a Run

0

0

0

Control and Columns

Columns · Treat as

Control · Status

WHAT THE LAST RUN FOUND

Pockets worse and material, charge-offs

Dollars above their share, in those

Pockets keeping less, worse and material

4 of 81

\$3,094,991

4 short \$3,248,654

Largest, worse and material	Segment	Loans	× its comparison	Dollars above share
FICO 496 - 653	Broker	523	2.62×	\$1,494,129
FICO 496 - 653	ASSET_CLASS 4	392	1.74×	\$643,768
ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	1.75×	\$529,198
FICO 712 - 745	Online	564	2.06×	\$427,896

[Every pocket, every measure: the Pockets tab.](#)

THE TABS

You answer

Results

Record

[Control: the professional calls](#)

[Pockets: every pocket, worse first](#)

[Record: what ran, the tie-outs, every Run](#)

[Columns: meanings, odd values, memory](#)

[Paid, cost, kept: paid against cost](#)

[Look: each number column's shape](#)

[Grids: one grid at a time, and how common](#)

[Split: each pocket halved](#)

Correct screen: Where things stand is all zeros: no answers needed before Run, no odd value left, no change waiting. **What the last Run found** gives the same numbers as the window, then the largest pockets that are worse and material. On the walk the top one was FICO 496 to 653 / Broker, 2.62 times its band's charge-offs, \$1,494,129 above its share. That is the pocket planted in the made-up book.

Step 14 · Read Pockets

Every pocket, worst first. Pick the **Measure** (Bad loans, Bad dollars, Charge-offs, Kept after losses, Earned before losses), **Pockets** (two-way, or split) and **Show** at the top.

Pockets

Which pockets lose more than their share, how much, and whether it's real and big enough to matter.

How this tab works

A pocket

One band of one column crossed with one segment of another, in every grid chosen in the launcher. Loans: how many are in it. Split by REV_DEBT: each pocket halved at its own median REV_DEBT, each half tested like any other pocket.

Rest of band

Every other loan in the same band, judged against on Control picks it or the rest of the book (every other loan in the book); now the rest of its band. It decides the gap, the excess and both verdicts. A pocket alone in its band has nothing beside it to compare with, so it is judged against the rest of the book, whatever Control says.

Excess

What the pocket lost above what it would have lost at the rest's rate: bad loans for Bad loans, dollars for the dollar measures. For Kept after losses and Earned before losses, the dollars it fell short of the rest, and the gap is in points of booked dollars.

Worse?

Yes: at least 1.34x the rest (for profit, short of it by the profit line on Control), with a p-value under 5%. Not sure: the gap is there, but the p-value isn't under it. Too few losses: fewer than 10 (a setting for the next Run), so not tested. No: anything else.

p-value

The chance of a gap at least this big if the pocket were no different from the rest, after the allowance for testing many pockets at once (Benjamini-Hochberg, across one grid and one measure). Bad loans: the z test, or the exact test for a pocket under 65 loans. Dollar measures: the loans are shuffled 10,000 times, within the band for the rest of its band, and it is how often a gap as big turned up. Under 5% counts.

Material?

Yes when the excess reaches the materiality line on Control (the Material at tile, in the measure's own unit). It is judged apart from Worse?: a pocket can be one without the other.

Could have caught

The smallest gap a pocket this size would catch 80% of the time at the last Run's confidence. As of the last Run.

Order

Sorted as of the last Run: worse pockets first, largest excess first, then the rest by their excess. Last come pockets losing more only against the other comparison, so that changing judged against on Control finds them here. A change on Control updates Worse?, Material?, the dollars and the colours at once, not the order. The caption counts what the dropdowns pick.

Holds FICO fixed?

Split only. REV_DEBT moves with FICO (correlation -0.53), so a split pocket in a grid that doesn't hold FICO fixed may be showing FICO, not REV_DEBT. Those rows come last, in grey, with no verdict colour.

LINES IN USE NOW

Live from Control

Worse at

1.34x

Better at

0.75x

How sure

95%

Material at

6.2 bad loans

Judged against

Rest of its band

Order and "Could have caught" are from the last Run, 2026-09-27 08:23.

MEASURE

POCKETS

SHOW

Bad loans

Two-way

All

7 worse and material · 7 worse · 41 shown

#	Band	Segment	Loans	This pocket	Rest of band	× rest of band	Bad loans above share	Worse?	p-value	Material?	Could have caught
1	FICO 496 - 653	Broker	523	24.47%	10.27%	2.38×	74.3	Yes	under 0.01%	Yes	1.46×
2	FICO 712 - 745	Online	564	8.69%	3.63%	2.39×	28.5	Yes	0.02%	Yes	1.45×
3	FICO 496 - 653	ASSET_CLASS 4	392	20.41%	13.24%	1.54×	28.1	Yes	1.4%	Yes	1.54×
4	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	12.32%	7.00%	1.76×	22.0	Yes	1.5%	Yes	1.52×
5	FICO 654 - 685	ASSET_CLASS 4	410	11.71%	7.26%	1.61×	18.2	Yes	4.5%	Yes	1.52×
6	ORIG_BAL 26,324 - 37,950	ASSET_CLASS 4	412	9.95%	5.82%	1.71×	17.0	Yes	4.3%	Yes	1.52×
7	FICO 712 - 745	ASSET_CLASS 4	427	8.20%	4.39%	1.87×	16.2	Yes	3.4%	Yes	1.51×
8	ORIG_BAL 15,449 - 26,323	Broker	537	11.73%	8.09%	1.45×	19.6	Not sure	11.0%	Yes	1.46×
9	ORIG_BAL 49,152 - 59,984	Broker	520	8.46%	5.46%	1.55×	15.6	Not sure	11.0%	Yes	1.47×
10	FICO 686 - 711	Online	546	7.69%	5.31%	1.45×	13.0	Not sure	16.0%	Yes	1.46×
11	ORIG_BAL 15,449 - 26,323	ASSET_CLASS 4	413	11.38%	8.59%	1.32×	11.5	No	23.3%	Yes	1.52×
12	ORIG_BAL 5,005 - 15,448	ASSET_CLASS 4	425	9.65%	6.98%	1.38×	11.3	Not sure	21.9%	Yes	1.51×
13	ORIG_BAL 49,152 - 59,984	ASSET_CLASS 4	384	8.59%	5.76%	1.49×	10.9	Not sure	19.3%	Yes	1.54×
14	FICO 686 - 711	ASSET_CLASS 4	407	8.11%	5.45%	1.49×	10.8	Not sure	16.6%	Yes	1.53×
15	ORIG_BAL 37,951 - 49,151	Broker	531	9.23%	7.95%	1.16×	6.8	No	86.7%	Yes	1.46×
FICO 712 - 745											
1	FICO 712 - 745	ASSET_CLASS 4	427	8.20%	4.39%	1.87×	16.2	Yes	3.4%	Yes	1.51×

#

Band

Segment

Loans

This pocket

Rest of band

× rest of band

Bad loans above share

Worse?

p-value

Material?

Could have caught

1	FICO 496 - 653	Broker	523	24.47%	10.27%	2.38×	74.3	Yes	under 0.01%	Yes	1.46×
2	FICO 712 - 745	Online	564	8.69%	3.63%	2.39×	28.5	Yes	0.02%	Yes	1.45×
3	FICO 496 - 653	ASSET_CLASS 4	392	20.41%	13.24%	1.54×	28.1	Yes	1.4%	Yes	1.54×
4	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	12.32%	7.00%	1.76×	22.0	Yes	1.5%	Yes	1.52×
5	FICO 654 - 685	ASSET_CLASS 4	410	11.71%	7.26%	1.61×	18.2	Yes	4.5%	Yes	1.52×
6	ORIG_BAL 26,324 - 37,950	ASSET_CLASS 4	412	9.95%	5.82%	1.71×	17.0	Yes	4.3%	Yes	1.52×

Correct screen: the **Lines in use now** bar repeats your Control answers. **Worse?** is Yes, Not sure (a gap, but it could be chance) or No. **Material?** is Yes when the excess reaches your dollar line.

Step 15 • Read Paid, cost, kept

What each pocket paid the bank, what its losses cost, and what was kept. **Together** reads both sides at once: Net drain is more charge-offs and less kept; Strong is fewer charge-offs and more kept; Losing more, profit holding is more charge-offs while what was kept is about the same.

FICO x CHANNEL									
Band	Segment	Paid us - gap vs band			Cost us - charge-offs			Kept - gap vs band	
		Loans	Gap pts	Dollars	x band	Dollars	Gap pts	Dollars	Together
496 - 653	Broker	523	+0.91 pts	154,274	2.62x	1,494,129	-7.86 pts	-1,339,855	Net drain
712 - 745	Online	564	+4.44 pts	815,246	2.06x	427,896	+2.10 pts	386,053	Losing more, profit holding
496 - 653	Branch	502	-0.40 pts	-65,750	0.55x	-714,057	+3.97 pts	648,307	Strong
496 - 653	Online	531	-0.51 pts	-88,007	0.55x	-772,690	+3.94 pts	684,683	Strong
686 - 711	Online	546	+5.72 pts	1,010,351	1.71x	318,288	+3.92 pts	692,063	
654 - 685	Branch	503	-1.18 pts	-190,133	1.17x	116,497	-1.90 pts	-306,630	
746 - 921	Branch	518	+0.15 pts	24,877	1.26x	77,733	-0.32 pts	-52,856	
746 - 921	Broker	536	+0.13 pts	22,118	0.90x	-35,929	+0.34 pts	58,047	
654 - 685	Online	505	+2.49 pts	395,179	0.95x	-39,945	+2.74 pts	435,124	
746 - 921	Online	517	-0.28 pts	-47,229	0.87x	-43,146	-0.02 pts	-4,083	
654 - 685	Broker	531	-1.25 pts	-216,386	0.90x	-78,887	-0.79 pts	-137,499	
686 - 711	Broker	501	-2.98 pts	-492,248	0.78x	-124,809	-2.22 pts	-367,438	
712 - 745	Branch	538	-2.60 pts	-428,484	0.68x	-176,741	-1.51 pts	-249,929	
686 - 711	Branch	515	-2.90 pts	-489,194	0.69x	-184,658	-1.81 pts	-304,536	
712 - 745	Broker	509	-2.09 pts	-339,715	0.60x	-226,126	-0.71 pts	-114,483	
(marked missing)	Broker	54	-0.42 pts	-7,061	3.93x	64,811	-4.31 pts	-71,872	
(blank)	Broker	1	-2.93 pts	-820	0.00x	-1,165	+1.23 pts	345	
(marked missing)	Online	55	+4.23 pts	78,126	0.63x	-19,915	+5.30 pts	98,041	
(marked missing)	Branch	51	-3.90 pts	-69,133	0.23x	-47,224	-1.24 pts	-21,908	

Correct screen: FICO 496 to 653 / Broker reads **Net drain**, and FICO 712 to 745 / Online **Losing more, profit holding**. The chart puts each pocket by its charge-offs (across) and what it kept (up).

Step 16 • Read Grids

One grid at a time: the rate in each pocket, the rate against the whole book, against the rest of its band, and how many loans each pocket holds. Pick the grid and measure at the top.

Grids

One grid at a time: the rate, how it compares, and how many loans within each pocket.

How this tab works

Rate

The measure picked above, in every pocket: one band (down) by one segment (across), with the band's and the segment's totals in All.

vs the book

The pocket's rate over the book's. 2.00x goes bad twice as often as the book. Kept after losses and Earned before losses are a gap in points instead.

vs rest of band

The pocket's rate over every other loan in its band, so a band that is bad all through doesn't make each of its pockets look bad.

Colour

Red is worse, green better, pale in line: 0.5x and under the greenest, 2x and over the deepest red; for a gap in points, the largest in the grid is the deepest. A blank: fewer losses than fewest losses on Control, so not compared (a setting for the next Run).

Loans

How many loans are in each pocket, shaded by its share of the grid: darker is more. A count, not a test, so no red or green.

Groups

Under the blocks: how many loans, and booked dollars, fall in each group of the split column or a new column, pocket by pocket. The count is checked against the grid's before it is shown.

As of

Everything here is as of the last Run: no line on Control changes it.

GRID

FICO x CHANNEL

MEASURE

Bad loans

Rate - Bad loans

FICO	Branch	Broker	Online	All
496 - 653	10.18%	24.47%	10.34%	13.09%
654 - 685	8.95%	8.10%	8.32%	8.45%
686 - 711	5.44%	5.19%	7.69%	6.15%
712 - 745	4.09%	3.14%	8.69%	5.40%
746 - 921	4.44%	3.79%	3.68%	3.95%
(blank)		0.00%		0.00%
(marked missing)	1.96%	12.96%	3.64%	6.25%
All	6.47%	9.04%	7.69%	7.74%

vs the book

FICO	Branch	Broker	Online	All
496 - 653	1.32x	3.85x	1.34x	1.94x
654 - 685	1.16x	1.05x	1.07x	1.09x
686 - 711	0.70x	0.67x	0.99x	0.79x
712 - 745	0.53x	0.43x	1.12x	0.70x
746 - 921	0.57x	0.48x	0.47x	0.51x
(blank)				
(marked missing)	0.84x	1.17x	0.99x	1.00x
All				

Loans

FICO	Branch	Broker	Online	All
496 - 653	502	523	531	1,556
654 - 685	503	531	505	1,539
686 - 711	515	501	546	1,562
712 - 745	538	509	564	1,611
746 - 921	518	536	517	1,571
(blank)				
(marked missing)	51	54	55	160
All	2,627	2,655	2,718	8,000

vs rest of band

FICO	Branch	Broker	Online	All
496 - 653	0.59x	2.38x	0.59x	
654 - 685	1.09x	0.94x	0.98x	
686 - 711	0.84x	0.79x	1.45x	
712 - 745	0.68x	0.49x	2.39x	
746 - 921	1.20x	0.92x	0.90x	
(blank)				
(marked missing)				
All				

A blank: fewer losses than the minimum, so not compared.

Step 17 • Read Split

Every pocket cut in half at its own middle REV_DEBT. Does the high half do worse?

Split

Every pocket split in two at its own median REV_DEBT. does the high half do worse?

How this tab works

What it does

Inside each pocket the loans are sorted by REV_DEBT and cut at that pocket's own median. The high half is compared with the low half, so the band and segment are the same on both sides. REV_DEBT isn't cut into bands of its own while it splits.

High vs low

The high half's rate over the low half's: 2.00x means the high half goes bad, or loses, twice as often. Kept after losses and Earned before losses are a gap in points instead: +0.30 pts means the high half keeps 0.30 points more per booked dollar.

High vs low, all

The high halves' actual total against what it would be at their low halves' rates, added over every pocket, with its range at 95% sure. For Bad loans only, the odds are pooled too (Mantel-Haenszel: a standard way to combine pockets without mixing their loans), with their p-value (Cochran-Mantel-Haenszel), and Cochran's Q asks whether the gap is the same size in every pocket.

p-value

The chance of a gap at least this big if the two halves were no different. Under 5% counts, and is bold. Bad loans: the gap in standard errors; a dollar measure: the loans dealt into the two halves at random inside their pocket, 10,000 times. The pockets are after the allowance for many tests (Benjamini-Hochberg, across one grid and one measure); a gap that is not significant is in brackets, unshaded. The summary is one pooled test per grid and measure, with no allowance. A blank: a half has fewer loans or losses than the minimums (65 loans, 10 losses).

What it holds

A grid holds fixed only its band and segment. REV_DEBT moves with FICO (correlation -0.53), so in a grid that doesn't hold FICO fixed part of every gap may be FICO, not REV_DEBT: the chip says which. Grids that hold it fixed come first in the list. Loans are treated as independent of each other.

As of

The numbers are the last Run's. The range, the brackets and the bold follow the confidence on Control.

LINES IN USE NOW

live from Control

Worse at

Better at

How sure

Material at

Judged against

The numbers are from the last Run, 2026-09-27 08:23.

1.34x

0.75x

95%

\$107,354

Rest of its band

GRID

FICO x CHANNEL

Holds FICO fixed

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02).

Measure	Pockets	High half worse in	High vs low, all	Range (95% sure)	p-value	As odds	p-value, as odds
Bad loans	15	14 of 15	2.02x	1.79x to 2.25x	under 0.01%	2.19x	under 0.01%
Bad dollars	15	15 of 15	2.14x	1.87x to 2.40x	under 0.01%		
Charge-offs	15	14 of 15	2.13x	1.86x to 2.41x	under 0.01%		
Kept after losses	15	13 of 15	-2.78 pts	-3.65 to -1.92 pts	under 0.01%		
Earned before losses	15	5 of 15	+0.26 pts	-0.21 to +0.72 pts	29.2%		

Same in every pocket? Bad loans: no sign the gap differs between pockets; Bad dollars, Charge-offs, Kept after losses, Earned before losses: not tested: dollar rate.

MEASURE

Bad loans

Bad loans, high vs low

	Branch	Broker	Online
496 - 653	2.39x	2.67x	(1.63x)
654 - 685	(1.37x)	2.92x	2.24x
686 - 711	(2.12x)	(1.90x)	3.20x
712 - 745	(2.14x)	(0.78x)	(1.58x)
746 - 921	(1.30x)	(2.33x)	(1.12x)
(blank)			
(marked missing)			

p-value per pocket

	Branch	Broker	Online
496 - 653	0.76%	under 0.01%	12.8%
654 - 685	33.5%	0.40%	2.9%
686 - 711	12.7%	14.3%	0.31%
712 - 745	13.6%	66.1%	14.3%
746 - 921	60.3%	12.8%	80.9%
(blank)			
(marked missing)			

Correct screen: on Bad loans, the high half is worse in 14 of 15 pockets, 2.02 times overall. The grid holds FICO fixed, which matters because REV_DEBT moves with FICO. The tab says so beside the grid.

Step 18 · Read Record

What ran, whether it adds up, what was left out, and every Run. This is the tab for a reviewer.

Correct screen: without pressing Run, Start here now says **2 of 81** (it said 4), and its list of the largest keeps only the two still worse: FICO 496 to 653 / Broker and FICO 712 to 745 / Online. Pockets' **Worse?** column follows. Rows now under 2 times read **No**:

Live from Control		2.00x	0.75x	95%	6.2 bad loans	Rest of its band					
MEASURE		POCKETS		SHOW							
Bad loans		Two-way	All	2 worse and material - 2 worse - 41 shown							
#	Band	Segment	Loans	This pocket	Rest of band	x rest of band	Bad loans above share	Worse?	p-value	Material?	Could have caught
1	FICO 496 - 653	Broker	523	24.47%	10.27%	2.38x	74.3	Yes	under 0.01%	Yes	1.46x
2	FICO 712 - 745	Online	564	8.69%	3.63%	2.39x	28.5	Yes	0.02%	Yes	1.45x
3	FICO 496 - 653	ASSET_CLASS 4	392	20.41%	13.24%	1.54x	28.1	No	1.4%	Yes	1.54x
4	ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	12.32%	7.00%	1.76x	22.0	No	1.5%	Yes	1.52x
5	FICO 654 - 685	ASSET_CLASS 4	410	11.71%	7.26%	1.61x	18.2	No	4.5%	Yes	1.52x
6	ORIG_BAL 26,324 - 37,950	ASSET_CLASS 4	412	9.95%	5.82%	1.71x	17.0	No	4.3%	Yes	1.52x
7	FICO 712 - 745	ASSET_CLASS 4	427	8.20%	4.39%	1.87x	16.2	No	3.4%	Yes	1.51x
8	ORIG_BAL 15,449 - 26,323	Broker	537	11.73%	8.09%	1.45x	19.6	No	11.0%	Yes	1.46x
9	ORIG_BAL 49,152 - 59,984	Broker	520	8.46%	5.46%	1.55x	15.6	No	11.0%	Yes	1.47x

Step 20 · Change a "Needs a Run" answer and see it wait

On Control, change **Fewest loans in a pocket for the usual test** to **100 loans**. Save.

PocketBook
Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 12:10

How this tab works

Where things stand What is left before Run, counted live from Control and Columns as you fill them in.

What the last Run found Pockets worse and material at Control's lines, and the five largest.

The tabs Red tabs you fill in; black tabs hold results; grey tabs are the record.

WHERE THINGS STAND

Answers needed before Run

0

0

1

Changes waiting for a Run

Control and Columns

Columns - Treat as

Control - Status

1 change is waiting for a Run. Fewest loans in a pocket for the usual test (smaller ones get the exact test): Enough for 5 expected losses (suggested) → 100 loans (Control C24). The result tabs still show the last Run. Save, close, and press Run in the launcher.

WHAT THE LAST RUN FOUND

Pockets worse and material, charge-offs

4 of 81

Dollars above their share, in those

\$3,094,991

Pockets keeping less, worse and material

4 short \$3,248,654

Largest, worse and material	Segment	Loans	x its comparison	Dollars above share
FICO 496 - 653	Broker	523	2.62x	\$1,494,129
FICO 496 - 653	ASSET_CLASS 4	392	1.74x	\$643,768
ORIG_BAL 37,951 - 49,151	ASSET_CLASS 4	414	1.75x	\$529,198
FICO 712 - 745	Online	564	2.06x	\$427,896

Every pocket, every measure: the Pockets tab.

Correct screen: Start here's **Changes waiting for a Run** says 1, in red, and a pink line names the change and its cell. On Control its **Status** reads **Waiting for a Run**. The choices made in the launcher, at the foot of Control, have a Status too, and read **Same as last Run**:

How far from the pocket's own test (suggested)

95% sure

1% of the book's total losses

The rest of its band

107,354

95% sure

1% of the book's total losses

The rest of its band

Needs a Run - takes effect when you press Run again

Setting

Fewest loans in a pocket for the usual test (smaller ones get the exact test)

100 loans

10 losses

5 bands (recommended)

Each band holds about the same number of loans (recommended)

80% of the time (recommended)

Hold down the share of false finds (Benjamin-Hochberg) (recommended)

On your own

65 (worked out from this book)

10 losses

5 bands

Each band holds about the same number of loans

80% of the time

Hold down the share of false finds (Benjamin-Hochberg)

Changes

1 Waiting for a Run

1 suggested: 65, from this extract at the last Run

Monitored and from the loans

Chosen in the launcher - read-only here

What are you running?

Where the book bleeds

FICO, ORIG_BAL

Segment by

CHANNEL, ASSET_CLASS

Split every pocket by

REV_DEBT

A number column with this many values or fewer is a category

12 values

A text column with more values than this is too fine to cut by

50 values

Same as last Run

Same as last Run

Same as last Run

Same as last Run

Same as last Run

Same as last Run

The result tabs still show the last Run until you press Run. Each says "1 Control change waits for a Run" beside **Lines in use now**.

Step 21 · Close the workbook and press Run again

PocketBook

✓ **Extract**
Consumer book Q3.csv ·
8,000 loans

✓ **Set up**
10 columns read

✓ **Choose tests**
Where the book bleeds

✓ **Answer in workbook**
All answered

✓ **Run**
12:10 · 9 seconds

Run finished. Results are in the workbook.

Pockets worse and
material

4
charge-offs, of 81

Charge-offs above their
share

\$3.09M
in those 4 pockets

Open Consumer book Q3 - PocketBook.xlsx: start with Pockets.
Split by REV_DEBT: each pocket halved at its own median. See the Split tab, and
Pockets split by REV_DEBT. REV_DEBT isn't cut on its own while it splits.

Run again

Open at Start here

Correct screen: "Run finished." The waiting count on Start here is back to 0.

Part C · Test new variables

A **new variable** is a column you think might tell good loans from bad, beyond what is already known. Here, revolving debt, with the FICO score held fixed so that REV_DEBT isn't just standing in for FICO.

Step 22 · Go back to Choose tests and pick Test new variables

Press **Choose tests** on the left, then **Test new variables** at the top right.

page 20 of 30

PocketBook

- ✓ **Extract**
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**
10 columns read
- 3 Choose tests**
Test new variables
- 4 Answer in workbook
- 5 Run

What are you running?

Where the book bleeds **Test new variables**

Column	What it is	Outcome	Test it	Hold fixed
LOAN_NBR	Loan number (key)			
FICO	FICO score		☐	☐
CHANNEL	Category · 3 values		☐	☐
ORIG_BAL	Booked amount		☐	☐
BAD_FLAG	Outcome (yes/no)	●		
GCO_AMT	GCO dollars			
RANR_AMT	RANR dollars			
ASSET_CLASS	Category · 4 values		☐	☐
REV_DEBT	Amount or number		☐	☐
ORIG_DATE	Origination date			

Find on 70% confirm on the rest
Or confirm a saved shortlist **Browse...**

Install scikit-learn

One add-on is missing: scikit-learn, which finds new variables (scouting). Press Install scikit-learn, or confirm a saved shortlist instead.

GCO dollars: what a loan charged off, in dollars.
RANR dollars: what we kept from a loan after its losses, in dollars.
Scouting: a first look at most of the loans, to pick what to test.

Saved to Control, read-only.

Next: answer in the workbook →

Correct screen: the columns now read **Outcome**, **Test it** and **Hold fixed**. If a red button says **Install scikit-learn**, press it. **scikit-learn** is the free add-on that does the scouting.

Step 23 · Wait for scikit-learn

PocketBook

- ✓ **Extract**
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**
10 columns read
- 3 Choose tests**
Test new variables
- 4 Answer in workbook
- 5 Run

What are you running?

Where the book bleeds **Test new variables**

Column	What it is	Outcome	Test it	Hold fixed
LOAN_NBR	Loan number (key)			
FICO	FICO score		☐	☐
CHANNEL	Category · 3 values		☐	☐
ORIG_BAL	Booked amount		☐	☐
BAD_FLAG	Outcome (yes/no)			
GCO_AMT	GCO dollars			
RANR_AMT	RANR dollars			
ASSET_CLASS	Category · 4 values		☐	☐
REV_DEBT	Amount or number		☐	☐
ORIG_DATE	Origination date			

Find on **70%** confirm on the rest Or confirm a saved shortlist **Browse...**

Tick at least one input to test.
GCO dollars: what a loan charged off, in dollars.
RANR dollars: what we kept from a loan after its losses, in dollars.

Installed scikit-learn. Everything PocketBook needs is here.

Saved to Control, read-only. **Next: answer in the workbook →**

Correct screen: a line says it was installed. The box under the table asks you to tick at least one input.

Step 24 · Tick what to test and what to hold fixed

Tick **Test it** on REV_DEBT, ASSET_CLASS, CHANNEL and ORIG_BAL. Tick **Hold fixed** on FICO. Leave **Find on** at **70%**: the first 70% of the loans by date are used to find, the rest to confirm.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**
10 columns read
- 3 Choose tests**
Test new variables
- 4 Answer in workbook
- 5 Run

What are you running?

Where the book bleeds
Test new variables

Column	What it is	Outcome	Test it	Hold fixed
LOAN_NBR	Loan number (key)			
FICO	FICO score		☐	☒
CHANNEL	Category - 3 values		☒	☐
ORIG_BAL	Booked amount		☒	☐
BAD_FLAG	Outcome (yes/no)	🎯		
GCO_AMT	GCO dollars			
RANR_AMT	RANR dollars			
ASSET_CLASS	Category - 4 values		☒	☐
REV_DEBT	Amount or number		☒	☐
ORIG_DATE	Origination date			

Find on 70% confirm on the rest
Or confirm a saved shortlist
Browse...

This will run: 4 inputs (REV_DEBT, ASSET_CLASS, CHANNEL, ORIG_BAL) against BAD_FLAG, each with and without FICO held fixed: 8 tests, found on 70% and confirmed on 30%.

GCO dollars: what a loan charged off, in dollars.
RANR dollars: what we kept from a loan after its losses, in dollars.

Installed scikit-learn. Everything PocketBook needs is here.

Saved to Control, read-only.
Next: answer in the workbook →

Correct screen: 4 inputs ... against BAD_FLAG, each with and without FICO held fixed: 8 tests, found on 70% and confirmed on 30%.

Step 25 · Press Next

PocketBook

✓ Extract

Consumer book Q3.csv · 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Test new variables

4 Answer in workbook

Fill in the shaded cells

5 Run

Answer in the workbook

Consumer book Q3 - PocketBook.xlsx is written beside the extract. Fill in the shaded cells, save and close it, then press Run.

Everything is answered. Press Run.

Open the workbook

Run

Correct screen: "Everything is answered. Press Run." Your Control answers carry over. Start here counts the new kind of run as **1** change waiting for a Run, because the result tabs still show the last one:

PocketBook

Consumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 12:10

How this tab works

Where things stand

What the last Run found

The tabs

What is left before Run, counted live from Control and Columns as you fill them in.

Pockets worse and material at Control's lines, and the five largest.

Red tabs you fill in; black tabs hold results; grey tabs are the record.

WHERE THINGS STAND

Answers needed before Run

0

Control and Columns

Odd values to answer

0

Columns · Treat as

Changes waiting for a Run

1

Control · Status

1 change is waiting for a Run. What are you running?: Where the book bleeds → Finding and testing a new variable (launcher). The result tabs still show the last Run. Save, close, and press Run in the launcher.

WHAT THE LAST RUN FOUND

Pockets worse and material, charge-offs

Dollars above their share, in those

Pockets keeping less, worse and material

Control now shows only the questions this kind of run uses, and **What are you running?** reads Finding and testing a new variable, **Waiting for a Run**:

How much worse than its comparison a pocket must be	The smallest significant gap in a typical pocket (suggested)	1.34x	1.34 (worked out from this book)	Worked out when
How sure a difference must be before it counts	95% sure		95% sure	
Smallest excess less worth reporting	1% of the book's total losses	\$107,354	1% of the book's total losses	

Needs a Run - takes effect when you press Run again				
Setting	Your answer to	On your own	Last Run used	Status
How many bands a number column is cut into	5 bands (recommended)		5 bands	Same as last Run
Where the band edges fall	Each band holds about the same number of loans (recommended)	N/A	Each band holds about the same number of loans	Same as last Run
Allowing for testing many pockets at once	Hold down the share of false finds (Benjamini-Hochberg) (recommended)	N/A	Hold down the share of false finds (Benjamini-Hochberg)	Same as last Run

Chosen in the launcher - read-only here		Finding and testing a new variable		To change these, go back to Choose tests in the launcher and press Next.	Waiting for a Run
What are you running?		Scout first			
Scout first, or test from a pre-spec already written?		FICO			
Cut into bands		None			
Segment by					
Split every pocket by					
Tested against		BAD_FLAG			
Inputs tested		REV_DEBT, ASSET_CLASS, CHANNEL, ORIG_BAL			
Held fixed		FICO			
Found on		70%, confirmed on the rest			
Saved shortlist (pre-spec file)					
A number column with this many values or fewer is a category		12 values			
A text column with more values than this is too fine to cut by		50 values			

Step 26 • Press Run

On 8,000 loans it took 20 seconds. PocketBook scouts on the development loans, writes the pre-spec Consumer book Q3 - pre-spec.yaml beside the workbook, and only then tests it on the held-back loans.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv - 8,000 loans
- ✓ **Set up**
10 columns read
- ✓ **Choose tests**
Test new variables
- ✓ **Answer in workbook**
All answered
- ✓ **Run**
12:11 • 21 seconds

Run finished. Results are in the workbook.

REV_DEBT groups worse than 11,000 - 14,999
1 of 2
on the holdout, 95% sure

Their share of bad loans
54%
on the holdout

Follows the pre-spec
Written now
by this Run's scouting

Open Consumer book Q3 - PocketBook.xlsx: start with New variables.
Ran on 8,000 loans from Consumer book Q3.csv. What was run: Finding and testing a new variable: scout first.

Run again

Open at Start here

Correct screen: the first tile names the column and its reference group (the middle group, which every other is compared with): **1 of 2** REV_DEBT groups worse, on the held-back loans. The last tile says the pre-spec was **written now, by this Run's scouting**. It can't differ from itself, so it doesn't say "Yes". Under the tiles: start with New variables.

Step 27 · Read Start here

PocketBookConsumer book Q3.csv · 8,000 loans · 10 columns · last Run 2026-09-27 12:11

How this tab works

Where things standWhat is left before Run, counted live from Control and Columns as you fill them in.

What the last Run foundEach group of the tested column on the holdout, against the reference group. Significant? follows the confidence on Control; the rest is as of the last Run.

The tabsRed tabs you fill in; black tabs hold results; grey tabs are the record.

WHERE THINGS STAND

Answers needed before Run	Odd values to answer	Changes waiting for a Run
0	0	0
Control and Columns	Columns · Treat as	Control · Status

WHAT THE LAST RUN FOUND

Groups worse than 11,000 - 14,999, on the holdout	Their share of the holdout's bad loans	Loans tested	Follows the pre-spec
1 of 2	54%	5,604 found · 2,395 held back	Written now: by this Run's scouting

Group of REV_DEBT	Loans	Bad rate	× 11,000 - 14,999's odds, FICO held fixed	p-value	Significant?	Share of bad loans
0 - 10,999	1,131	5.31%	1.24×	38.4%	No	32%
11,000 - 14,999 (reference)	546	4.76%	1.00×			14%
15,000 - 48,522	718	14.07%	2.69×	under 0.01%	Yes, worse	54%

[Found beside confirmed, with and without the columns held fixed, and how it's worked out: the New variables tab.](#)

Correct screen: one row per REV_DEBT group on the held-back loans. 15,000 and up is **Yes, worse:** 2.69 times the odds of going bad with FICO held fixed (the column heading says so), holding 54% of the bad loans.

Step 28 · Read Scouting

Which candidates the book leans on, ranked, on the development loans only.

ScoutingWhich of the 4 candidates does the book lean on, and where does each one bend? On the development loans only.

How this tab works

What it isScouting: which of the 4 candidates the book leans on, and where each one bends. It nominates; it never confirms (statistics.md B7). The confirmation is the New variables tab, on loans scouting never read.

Found on5,604 development loans made 2021-06-30 to 2024-10-24, 432 of them bad: the first 70% of the loans by origination date. The 2,395 loans made after them are held back. Only their dates were read here, to draw the line and to name the holdout's range; their outcomes and every other value were not. Left out: 1 with no readable outcome.

CandidatesThe columns ticked Test it in the launcher, and every new column made on Columns (one column divided by another), each fed in as a column in its own right. A category is given one number per value, in the order of its values.

The forestA random forest: 200 trees, each leaf at least 40 loans, seed 7, grown by scikit-learn 1.9.1. Each tree splits the loans wherever a cut best separates bad from good, and the forest averages their votes. It writes no formula and assumes no shape. The same extract gives the same forest; another version of scikit-learn may not.

ImportanceHow far the forest's AUC drops when the column is shuffled, on loans it did not train on. AUC is the chance a random bad loan scores above a random good one; 0.5 is a coin flip. The development loans are cut into 3 runs by date: each run is scored by a forest grown on the other two, with 5 shuffles of each column, and the drops are averaged. The forest's own AUC is 0.671. It is a ranking: no direction, no size and no range, and it leans towards columns with many distinct values.

With and withoutImportance is from a forest given the candidates alone. The next column is from a forest given FICO as well (AUC 0.689): a candidate that drops there was mostly telling you FICO. Their own importance is under the table.

Noise floor0.0255: the largest importance any candidate reached in the same forests grown on the development loans with their outcomes shuffled among them, so that no column could matter (20 tries). It is what a column scores by chance.

Proposed?Yes for a number column whose importance is above the noise floor, with the held-fixed columns in the forest or without them, and whose shape bends somewhere to cut. A category is ranked and not proposed: the pre-spec cuts a number at its bins.

Suggested binsFrom the shape: the forest's bad rate with the column set to one value for every loan (partial dependence, on 1,000 development loans drawn by seed), over the column's own percentiles, from a forest grown on every development loan with FICO in it. A cut goes where the curve steps by at least 25% of its average, largest step first, each group holding at least 2% of the loans, at most 6 groups. Inside each step the edge sits where the forest itself split the column most, rounded to two figures. The shapes are under the table.

ReferenceThe group holding the development loans' median value: the ordinary case, which every other group is compared with.

Correlated withCandidates, and held-fixed number columns, whose values rise and fall together with this one: a rank correlation (Spearman's) of 0.7 or more either way, on the development loans with both. Importance splits between such columns, and the shape of one can borrow the other's.

The pre-specPocketBook writes the pre-spec from the proposed candidates beside the workbook, when there is none, and the confirmation reads it. One that is there already is confirmed as it stands: an edit is yours to make, and Record logs it. The file is under the table.

As ofThe last Run, 2026-09-27 08:29. Nothing here follows Control until the next Run.

Outcome	Candidates · proposed	Held fixed	Noise floor	Development loans
BAD_FLAG	4 · 1 proposed	FICO	0.0255	5,604 · 432 bad

Worked out at the last Run, 2026-09-27 08:29, on the development loans. Ranked by the larger of the two importances.

Candidate	Rank	Importance	FICO held fixed	Suggested bins	Reference	Correlated with	Proposed?	Why
REV_DEBT	1	0.1540	0.0791	0 - 10,999; 11,000 - 14,999; 15,000 - 47,478	11,000 - 14,999		Yes	clears the noise floor
CHANNEL	2	0.0144	0.0075	Each value			No	a category
ASSET_CLASS	3	0.0080	0.0054	Each value			No	a category
ORIG_BAL	4	-0.0011	0.0030	5,024 - 6,199; 6,200 - 59,984	6,200 - 59,984		No	below the noise floor

Held fixed, for comparison: their own importance with the candidates

Column	Importance
FICO	0.0526

Correct screen: REV_DEBT first, **Proposed? Yes**, above the **noise floor** (what a column scores by pure chance). CHANNEL and ASSET_CLASS are categories, ranked but not proposed. ORIG_BAL is below the floor.

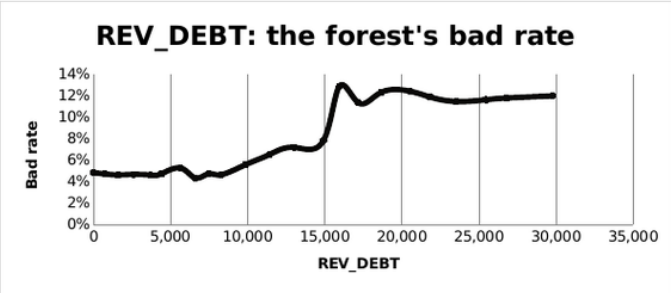
Further down, the pre-spec as written, and REV_DEBT's curve with the suggested cuts. Printed, the pre-spec's page has no table heading over it:

Written by this Run: Consumer book Q3 - pre-spec.yaml, beside the workbook, fingerprint b98435f2c897, before any held-back loan was tested. Edit it if you like; Record logs

```
# Written by PocketBook's scouting on 2026-09-27, from 5,604 development loans made 2021-06-30 to 2024-10-24.
# The 2,395 loans held back (made 2024-10-25 to 2026-03-30) were not read. Edit it if you like: Record logs every change.
prespec: 1
written: 2026-09-27
outcome: BAD_FLAG
inputs:
  - column: REV_DEBT
    bins: [11000, 15000]
    reference: 1          # 11,000 - 14,999, the group holding the median
strata: [FICO]
confidence: 0.95
holdout: {from: 2024-10-25, to: 2026-03-30}
development: {from: 2021-06-30, to: 2024-10-24}
```

REV_DEBT: the forest's bad rate as REV_DEBT moves (cut at 11000, 15000)

REV_DEBT	Bad rate
0	4.9%
733	4.7%
1,584	4.6%
2,599	4.7%
3,688	4.6%
4,406	4.8%
5,633	5.3%
6,584	4.4%
7,452	4.7%
8,282	4.7%
9,852	5.6%
11,392	6.5%
12,998	7.2%
14,907	7.8%
15,945	12.9%
17,188	11.4%
18,659	12.3%
20,560	12.4%
21,821	11.9%
23,510	11.5%
25,440	11.6%
26,778	11.8%
29,765	12.0%

REV_DEBT: the forest's bad rate

Step 29 · Read New variables

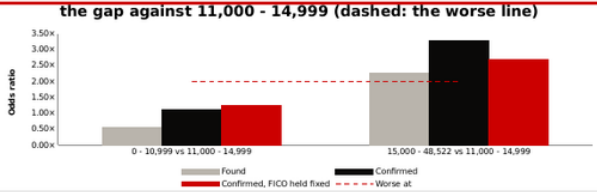
Does REV_DEBT still tell good loans from bad on loans it was never found on, and once FICO is held fixed?

WHAT WAS TESTED
from the shortest scouting write

Outcome	Candidates	Held fixed	Material at - live	Loans
BAD_FLAG	REV_DEBT	FICO	\$107,354	5,604 found - 2,395 held back

The numbers are from the last Run, 2026-09-27 08:29. The verdicts, the materiality line and the chart's worse line follow Control.

Candidate	Compared	Found - 5,604 loans		Confirmed - 2,395 held back		Confirmed, FICO held fixed		Excess \$	Material?	In words
		Gap	p. allowed	Gap	p. allowed	Gap	p. allowed			
REV_DEBT	0 - 10,999 vs 11,000 - 14,999	0.56x	under 0.01%	1.12x	63.7%	No	1.24x	18.4%	No	Didn't hold up on held-back loans
REV_DEBT	15,000 - 48,522 vs 11,000 - 14,999	2.26x	under 0.01%	3.27x	under 0.01%	Yes	2.69x	under 0.01%	Yes	Holds up, and not just FICO

the gap against 11,000 - 14,999 (dashed: the worse line)

Correct screen: 15,000 and up: **Holds up? Yes** and **Still holds? Yes** (with FICO held fixed), **Material? Yes**. Under 11,000: found on development, but it **didn't hold up** on the held-back loans.

Step 30 • Read Record

Record

What ran, whether it adds up, what was left out, and every Run. For the reviewer.

How this tab works

What it holds

Settings

Every Run

What the last Run was, whether it adds up, which tests it used, what it left out, and every Run and refusal. Nothing here is a result: the result tabs hold those.

Each Control setting: the answer in use now, and what the last Run used. A setting changed since the Run is shaded; the result tabs follow a Changes now setting at once, and a Needs a Run one at the next Run.

Newest first. Each entry is what that Run used, and a refusal says what it was waiting for.

This Run

Item	Detail
Extract	Consumer book Q3.csv
Loans run	8,000
What was run	Finding and testing a new variable: scout first
Record of this run	Consumer book Q3 - PocketBook - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Origination dates	2021-06-30 to 2026-03-30 (8,000 loans; 0 without a readable date)
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Scouting	4 candidates on 5,604 development loans made 2021-06-30 to 2024-10-24; 1 proposed: REV_DEBT. See the Scouting tab.
Scouting held back	2,395 loans made 2024-10-25 to 2026-03-30: none of their outcomes or values was read by scouting.
Scouting's pre-spec	Written by this Run: Consumer book Q3 - pre-spec.yaml, beside the workbook, fingerprint b98435f2c897, before any held-back loan was tested. Edit it if you like. Record logs every change.
Pre-spec	/tmp/claude-Qi/home-user-SATC643050fc-8769-5da1-b067-4db589c3c311/scratch/pad/walk27/book/Consumer book Q3 - pre-spec.yaml
Pre-spec commit	not in a git repository, so not committed: a pre-spec only counts once it is committed
Pre-spec fingerprint	b98435f2c897: the first 12 characters of the file's SHA-256, so any change to it shows
What the pre-spec says	written: 2026-09-27 outcome: BAD_FLAG inputs: 1 1. column: REV_DEBT 1. bins: 11,000, 15,000 (groups: 0 - 10,999; 11,000 - 14,999; 15,000 - 48,522) 1. reference: 11,000 - 14,999 strata: FICO confidence: 0.95 holdout: 2024-10-25 to 2026-03-30 development: 2021-06-30 to 2024-10-24
Warning	This run doesn't count as the pre-specified one until the pre-spec is committed, unchanged.
Differs from the pre-spec	nowhere: this run used what it says
Holdout	2,395 of this extract's loans were made in the holdout (2024-10-25 to 2026-03-30), the first on 2024-10-25 and the last on 2026-03-30. This run touched the holdout.
Confirmatory test	Development 2021-06-30 to 2024-10-24: 5,604 loans, 432 bad; holdout 2024-10-25 to 2026-03-30: 2,395 loans, 187 bad. See the New variables tab. Left out of this test only (the loan counts on Record include them): 1 with no readable outcome.
Runs that touched the holdout	1 on Record's Every Run, this one included
This pre-spec's held-back runs	In order: this run (b98435f2c897).

Does it add up

Check	Result
Bleed tabs	None: testing a new variable runs only the confirmatory test. Any left by an earlier Run were taken off.

Settings

Setting	To use now	Last Run used
How many bands a number column is cut into	5 bands	5 bands
Where the band edges fall	Each band holds about the same number of loans	Each band holds about the same number of loans
Smallest excess loss worth reporting	1.0% of the book's GCO, \$107,354	1.0% of the book's GCO, \$107,354
How much worse than its comparison a pocket must be	worse at 2.00x	worse at 2.00x
How sure a difference must be before it counts	95% sure	95% sure
Allowing for testing many pockets at once	Hold down the share of false finds (Benjamini-Hochberg)	Hold down the share of false finds (Benjamini-Hochberg)

Tests used

Where	Test
p-value	The chance of a gap at least this big if there were no real difference, after the allowance for many tests. Below 5% is significant, at 95% sure. Two-sided: a gap either way counts.

Correct screen: the pre-spec's fingerprint (a short code that changes if the file changes by even one character), and one run that touched the held-back loans.

Step 31 • Edit the pre-spec after the held-back run, and Run again

Open Consumer book Q3 - pre-spec.yaml in Notepad. Change bins: [11000, 15000] to bins: [10000, 16000]. Save. Press **Run again**.

This is what the pre-spec exists to catch: changing the test after seeing the answer. PocketBook runs it, and labels it.

PocketBook

✓ Extract

Consumer book Q3.csv - 8,000 loans

✓ Set up

10 columns read

✓ Choose tests

Test new variables

✓ Answer in workbook

All answered

✓ Run

12:11 · 19 seconds

Run finished. Results are in the workbook.

REV_DEBT groups worse than 10,000 - 15,999

1 of 2

on the holdout, 95% sure

Their share of bad loans

45%

on the holdout

Follows the pre-spec

Changed

after a held-back run: see Record

Open Consumer book Q3 - PocketBook.xlsx: start with New variables. Changed pre-spec: Consumer book Q3 - pre-spec.yaml changed after the held-back run of 2026-09-27 12:11: fingerprint b98435f2c897 then, 5fe791ff8df7 now. This run is on the changed pre-spec.

Run again

Open at Start here

Correct screen: the last tile says **Changed, after a held-back run**, in red, and the line under the tiles gives the date of the earlier run and both fingerprints. On Record, under **This Run**, a red **Warning** says the same:

	confidence: 0.99
	holdout: 2024-10-25 to 2026-03-30
	development: 2021-06-30 to 2024-10-24
Warning	This run doesn't count as the pre-specified one until the pre-spec is committed, unchanged.
Warning	Changed pre-spec: Consumer book Q3 - pre-spec.yaml changed after the held-back run of 2026-09-27 08:29: fingerprint b98435f2c897 then, 5fe791ff8df7 now. This run is on the changed pre-spec.
Differs from the pre-spec	nowhere: this run used what it says
Holdout	2,395 of this extract's loans were made in the holdout (2024-10-25 to 2026-03-30), the first on 2024-10-25 and the last on 2026-03-30. This run touched the holdout.
Confirmatory test	Development 2021-06-30 to 2024-10-24: 5,604 loans, 432 bad; holdout 2024-10-25 to 2026-03-30: 2,395 loans, 187 bad. See the New variables tab. Left out of this test only (the loan counts on Record include them): 1 with no readable outcome.
Runs that touched the holdout	2 on Record's Every Run, this one included
This pre-spec's held-back runs	In order: 2026-09-27 08:29 (b98435f2c897); this run (5fe791ff8df7, changed).

When PocketBook says no

What you did	What the window says	What to do
Opened it before the add-ons were installed	Black bar: "Three add-ons are missing ..." (Step 1)	Press Install now . If it fails, press Copy for IT and send them the note.

What you did	What the window says	What to do
Pressed Run before answering (Step 7)	"9 answers needed before Run", each with its cell	Press Open at on each, answer, save, close, Run.
Left the workbook open in Excel (Step 11)	Pink bar: "The workbook is open in Excel." Run is grey.	Save and close it. The bar goes by itself.
Picked Test new variables without scikit-learn (Step 22)	"One add-on is missing: scikit-learn ..."	Press Install scikit-learn , or confirm a saved shortlist instead.
The grids didn't add up (Step 32, below)	"Run stopped: the grids didn't add up to the book, so nothing was written."	Nothing you answered caused it. Send the file named on the screen.

Step 32 · What a stopped Run looks like

This can't be made to happen from the screens. On the walk it was forced, to see what the window says.

PocketBook

- ✓ **Extract**
Consumer book Q3.csv · 8,000 loans
- ✓ **Set up**
10 columns read
- ✓ **Choose tests**
Where the book bleeds
- ! **Answer in workbook**
Run stopped
- 5 **Run**

Run stopped

Nothing was written. Your answers so far are kept.

Run stopped: the grids didn't add up to the book, so nothing was written. grid fico x channel: rows adds up to 7999 across the cells, but the book says 8000.

That is a fault in PocketBook, not in your answers. The details are in
/tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-4db589c3cc31/scratchpad/walk27/home2/pocketbook/last-error.txt;
send that file over to get it fixed.

[Open the workbook](#)
[Run](#)

Correct screen: "Run stopped", not "answers needed". The message says the grids didn't add up to the book, that nothing was written, and where the details are.